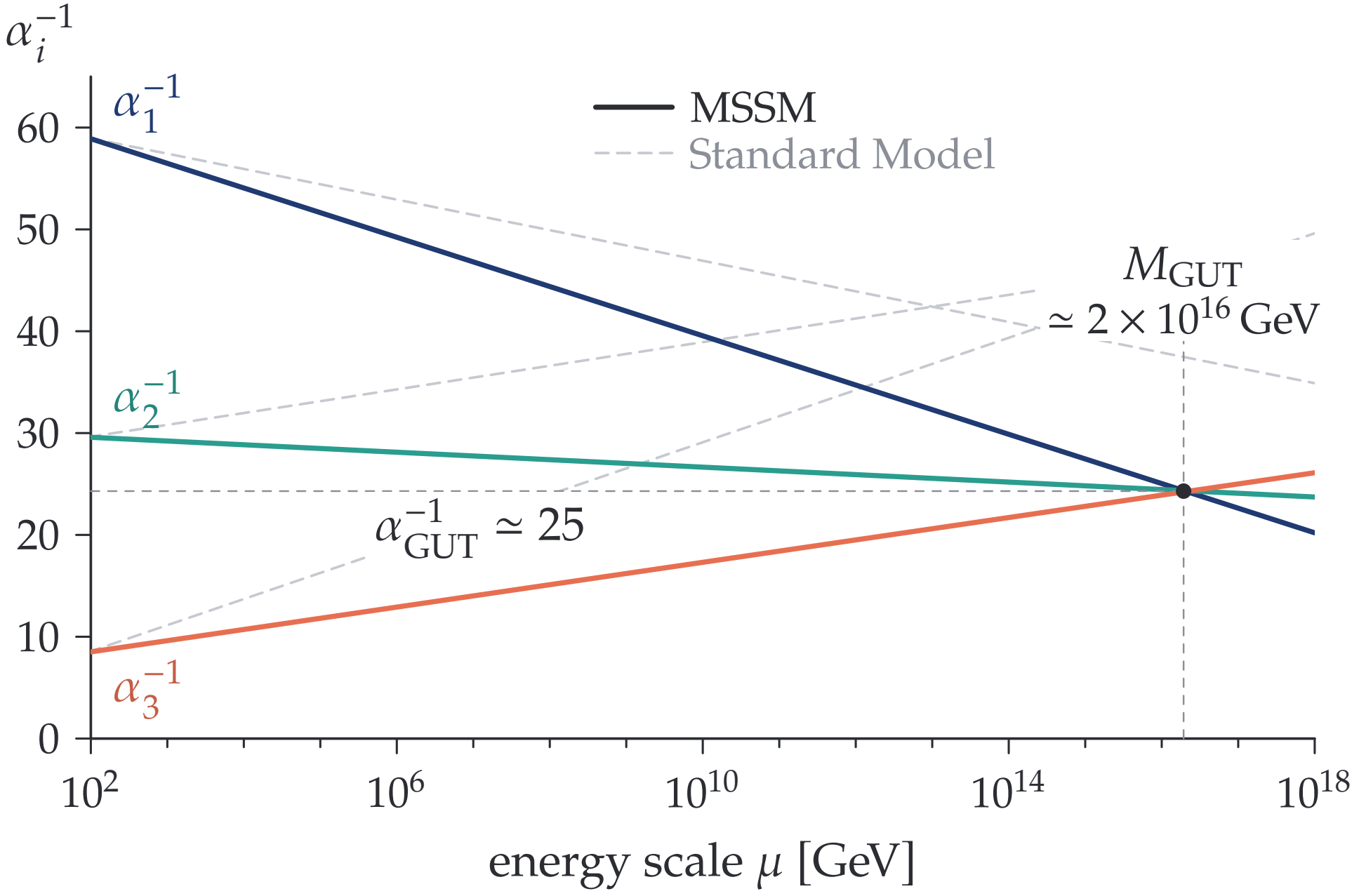
α_i^{-1}
α_1^{-1}
α_2^{-1}
α_3^{-1}
MSSM
Standard Model
M_{GUT}
$\simeq 2 \times 10^{16}$ GeV
$\alpha_{\text{GUT}}^{-1} \simeq 25$
60
50
40
30
20
10
0
10^2
10^6
10^{10}
10^{14}
10^{18}
energy scale μ [GeV]